

SMART AI EXPENSE TRACKER : A MOBILE – BASED INTELLIGENT FINANCIAL MANAGEMENT SYSTEM

Brief Description:

Smart AI Expense Tracker is a production-ready cross-platform mobile application built using React Native and Expo Go SDK 54. The application offers an intelligent solution for personal finance management by integrating three real AI features — an AI Financial Advisor powered by Claude API (Anthropic), a Smart Budget Prediction system using GPT-4o (OpenAI), and an automated Monthly Financial Report Card with letter grades and AI-written summaries. The backend is built on Node.js 20, Express 4, and MongoDB Atlas, deployed on Render cloud platform. Firebase Authentication secures all 17 screens. The system provides real-time expense tracking, budget alerts, spending predictions, and personalized financial advice — all backed by live data with zero mock content.

Principle:

The application employs Large Language Models (LLMs) — specifically Anthropic's Claude API and OpenAI's GPT-4o — to deliver intelligent financial guidance. Claude API uses Server-Sent Events (SSE) streaming to provide real-time conversational advice by injecting the user's last 90 days of actual expense data as context into every prompt. GPT-4o analyzes 3–6 months of historical spending to predict next month's budget per category with a confidence percentage, using structured JSON output validated by Zod schema. A node-cron scheduler auto-generates monthly report cards on the 1st of each month, computing letter grades (A+ to F) per spending category and an overall financial health score (0–100) based on savings rate, budget adherence, and expense streak. All predictions and advice are based on real user data — never hardcoded or mocked.

Beneficiaries:

- Students & Young Professionals
- Working Individuals & Salaried Employees
- Small Business Owners
- Low-Income Families
- Financial Literacy Educators

Benefits:

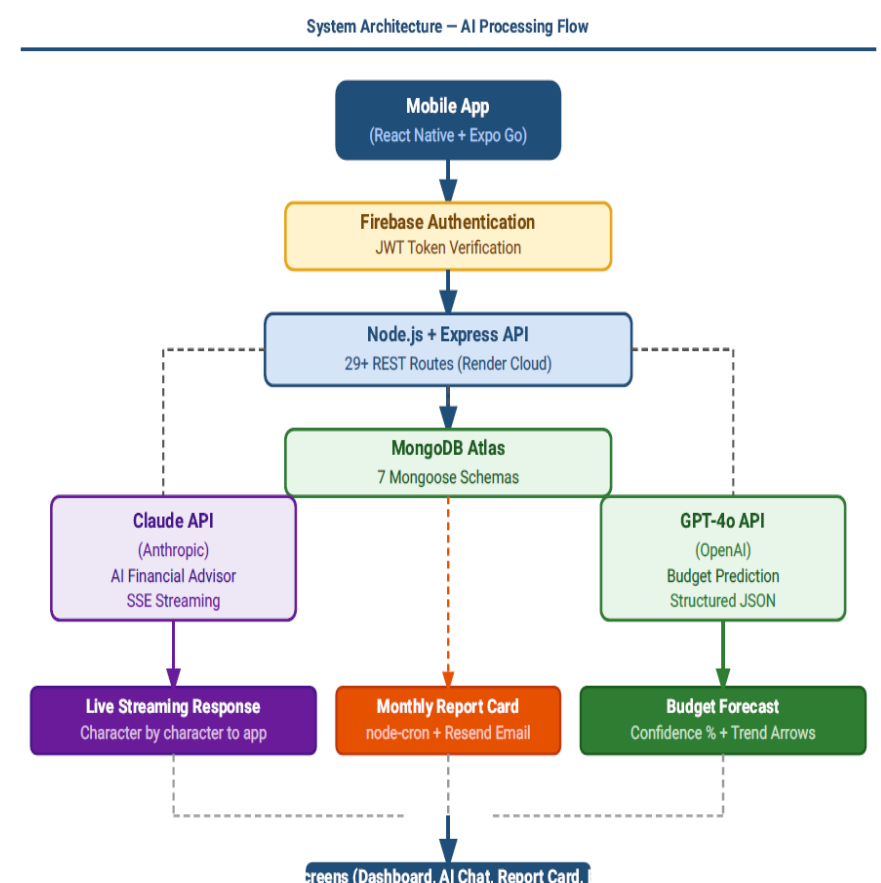
- Real-time AI-powered personalized financial advice
- Automated monthly budget predictions before month starts
- Reduced impulsive spending via instant budget alerts
- Improved financial literacy through graded report cards
- Democratized financial advisory — free & always available
- Promotes savings habit via goal tracking & progress rings

Technologies used:

- React Native + Expo Go SDK 54
- Node.js 20 + Express 4 + TypeScript
- MongoDB Atlas + Mongoose 8
- Claude API (Anthropic) + GPT-4o (OpenAI)
- Firebase Authentication
- Render (Deployment) + EAS Build (APK)

Features:

- Expense Tracking
- Budget Management
- Financial Analytics
- Multi-Currency Support
- AI Financial Advisor



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